

Youssef Handi

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Curriculum Vitae

Research Profile

I am a mathematical statistician and postdoctoral researcher at Western University, working at the intersection of mathematical statistics, statistical learning, copula-based dependence modelling, high-dimensional inference, spatial and spatio-temporal statistics, and stochastic processes. My research develops rigorous, interpretable, and computationally tractable methods for dependent, high-dimensional, imbalanced, non-Gaussian, and spatially structured data, as well as data involving rare events, with applications in health, reliability, environmental monitoring, risk management, and data science.

Current Position

Postdoctoral Researcher

2026–present

Western University, London, Ontario, Canada

- Developing finite-sample and high-dimensional inference methods for testing regression coefficients and block effects after adjustment for nuisance variables.
- Developing methods based on stochastic processes for occupation times, first excursions, recovery times, and resilience functionals.
- Establishing methodological links between high-dimensional inference, non-Gaussian dependence modelling, stochastic processes, and data science.

Research Interests

High-dimensional regression and block testing; copula-based dependence modelling and factor copulas; non-Gaussian spatial and spatio-temporal statistics; statistical learning and machine learning; semiparametric inference; stochastic processes; rare-event and risk modelling; robust and asymmetric regression; mixed discrete–continuous data; reproducible statistical computing.

Education

Ph.D. in Mathematics and Statistics

2021–2025

Université du Québec à Montréal (UQAM)

- Supervisors: Prof. Karim Oualkacha and Prof. Mhamed Mesfioui. Cumulative GPA: 4.00/4.3.
- Thesis: *Copula-based Regression Frameworks for Multivariate Binary Outcomes*.

Master's Degree in Computer Science and Applied Mathematics

2019–2021

Université du Québec à Trois-Rivières (UQTR)

- Supervisors: Prof. Mhamed Mesfioui and Prof. Georges Abdul-Nour. Cumulative GPA: 4.03/4.3.
- Thesis: *Combining Expert Opinions in Reliability using Bayesian Methods*.

Publications

- Handi, Y., Oualkacha, K., & Mesfioui, M. (2025). *On multivariate binary outcomes copulas-regression problem*. *Japanese Journal of Statistics and Data Science*. doi:10.1007/s42081-025-00322-0.
- Handi, Y., Oualkacha, K., & Mesfioui, M. (2026). *Inference based on copulas for quantile and expectile for binary regression*. *Journal of Statistical Theory and Practice*, 20, Article 61. doi:10.1007/s42519-026-00577-0.

Manuscripts in Progress and Current Projects

- Handi, Y., Ahmed, S. E., & Provost, S. B. *An Adaptive Haar Sparse-Spectral Test for Partial High-Dimensional Linear Regression*. Postdoctoral manuscript on testing identifiable residual block effects, sparse and spectral alternatives, and finite-sample calibration using Haar directions. The manuscript is 90% complete, with submission planned for September.
- Handi, Y., Provost, S. B., & Ahmed, S. E. *Non-Gaussian Local Factor Copulas on Complex Domains with Spatially Varying Loading Surfaces*. Manuscript on latent factor copulas for complex domains, local loading surfaces, scale-mixture dependence, composite-likelihood screening, active-set ECM/MCEM refitting, and tail theory. Submitted to the *Journal of Multivariate Analysis*; under review.

- Handi, Y., Provost, S. B., & Ahmed, S. E. *Geometry-Indexed Bernstein Vine Copulas for Non-Gaussian Spatial Data*. Manuscript on geometry-dependent Bernstein weight surfaces, exact copula constraints through Sinkhorn normalization, scalable vine factorizations, semiparametric estimation, and spatial prediction. In progress.
- Handi, Y. *Nonseparable Spatio-Temporal Geometry-Indexed Bernstein Vine Copulas*. Manuscript on dependence-shape nonseparability, centered space–time interactions, local Bernstein resolution, interaction-aware candidate screening, and one-step probabilistic prediction. In progress.
- Handi, Y. *Random Spatial Propagation Copulas: Identification of Graph Regimes and Extremal Geometry*. Manuscript using graph-structured random precision and correlation matrices to model changing spatial connectivity, with finite-regime identification, higher-order dependence, tail analysis, and probabilistic precipitation prediction. In progress.
- Handi, Y. *Geometry-Indexed Transport Copulas: Estimation, Prediction, and Structured Approximation*. Manuscript constructing spatial copulas through entropic multi-marginal optimal transport, with geometry-dependent rank-interaction costs, likelihood inference, graphical approximation, and conditional predictive-risk bounds. In progress.
- Handi, Y. Occupation-time, first-excursion, and recovery-time methods for jump-diffusion and Lévy-type processes, with applications to resilience, interruption durations, recovery, and risk in stochastic systems.
- Handi, Y., Oualkacha, K., & Mesfioui, M. Extension of copula-based regression for multivariate binary outcomes to mixed variables. Manuscript completed; being prepared for submission.
- Handi, Y., & Taamouti, A. Default probability under measurement error. Being prepared for submission.
- Handi, Y. Extremile regression based on copula dependence structures. In progress.

Research Experience

Doctoral Researcher

2021–2025

Statistical Modelling, UQAM

- Developed copula-based regression frameworks for multivariate binary outcomes.
- Developed copula-based quantile and expectile inference methods for binary regression.
- Extended dependence models to mixed continuous–discrete variables and nonstandard outcomes.
- Developed projects on extremiles, spatial dependence, complex domains, and rare events.

Research Assistant

2019–2021

Hydro-Québec Research Institute (IREQ), Montréal

- Developed Bayesian methods to combine expert opinions and observed machine performance data in reliability analysis.
- Implemented MCMC simulations to predict system failures.
- Used copula functions to model dependence in reliability engineering.

Teaching Experience

Lecturer

Summer 2025

Université du Québec à Montréal

- MAT-4681: Statistics for the Sciences, undergraduate level.
- Full responsibility for the syllabus, lectures, assignments, examinations, and student support.

Teaching Assistant

2022–2025

Université du Québec à Montréal

- MAT-1700: Probability; STT-2120: Regression Analysis; STT-1000: Statistics I.

Teaching Assistant

2019–2021

Université du Québec à Trois-Rivières

- STT-1030: Introduction to Regression; STT-1001: Probability Theory; STT-1040: Analysis of Variance; STT-1028: Time Series Analysis; MPU-1058: Measure and Integration.

Professional Experience

Data Scientist

2021–2024

National Bank of Canada, Montréal

- Designed machine learning algorithms for credit-risk classification for more than 20,000 credit card holders.
- Developed stress-testing models for adverse market scenarios.
- Built data ingestion pipelines using Kafka, JSON, and SQL sources, and implemented data quality checks for analytics platforms.

Scientific Presentations

- Invited talk, Université de Sherbrooke, Department of Mathematics, July 2025: *Multivariate Binary Outcomes Copulas-Regression Framework*.
- UQAM Statistics Seminar, Montréal, May 2025: *Multivariate Binary Outcomes Copulas-Regression Framework*.
- Francophone Interfaculty Research Colloquium in Biostatistics, Montréal, July 2024: *Advances in Copula Regression for Discrete Data*.

Scholarships and Awards

- Scholarship from the Fields Institute for Research in Mathematical Sciences, Toronto.
- Institut des sciences mathématiques Scholarship, 2022–2025. Awarded for academic excellence in mathematical research.
- Hydro-Québec Master's Research Scholarship, 2019–2021. Support for research on Bayesian methods in reliability.

Technical Skills

- **R:** VineCopula, rvinecopulib, copula, MASS, Matrix, RSpectra, RcppArmadillo, e1071, pracma; reproducible workflows with R Markdown and targets/drake.
- **Python:** NumPy, SciPy, scikit-learn, PyTorch, TensorFlow/Keras, PyTorch Lightning, copulas, vinecopulib, filterpy, pykalman, pyamg.
- **Scientific Computing and Workflows:** sparse linear algebra, simulation, parallel computing, GPU acceleration, Docker, Singularity, Snakemake, Make, GitHub Actions.

Languages

Arabic: native proficiency.

French: native proficiency.

English: professional proficiency in reading, writing, speaking, teaching, and scientific communication.

Academic References

Prof. Karim Oualkacha

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Université du Québec à Montréal (UQAM)
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Prof. Georges Abdul-Nour

Professor Emeritus, Department of Industrial Engineering
Université du Québec à Trois-Rivières (UQTR)
georges.abdulnour@uqtr.ca

Prof. S. Ejaz Ahmed

Distinguished Professor, Department of Mathematics and
Statistics
Brock University
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Prof. Mhamed Mesfioui

Full Professor, Department of Mathematics and
Computer Science
Université du Québec à Trois-Rivières (UQTR)
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Prof. Serge B. Provost

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